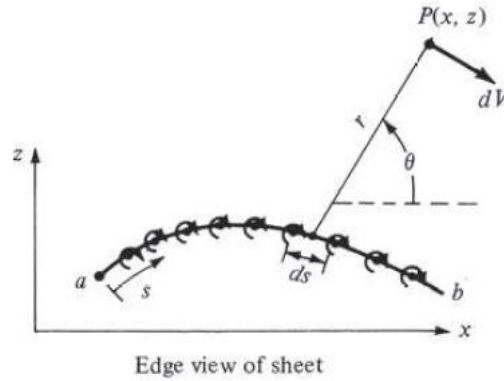


Vortex Panel Method

- This method does work for **lifting bodies** because the vortices add circulation.
- It is a much more widely used technique.



- Consider a vortex sheet on a curve **s** that starts at **a** and ends at **b**.
- The sheet has strength as a function of the s-coordinate.
- Strength for point vortex is represented by Γ .
- The strength parameter for the vortex sheet is represented as γ .
- Consider an arbitrary point **p** in space at a distance **r** from the segment **ds**, marking the azimuthal distance θ .
- The small section of the vortex sheet of strength γds induces an infinitesimally small velocity **dV** at point **p**.

$$dV = -\frac{\gamma ds}{2\pi r}$$

- The velocity potential at the arbitrary point **p** due to this segment **ds** is

$$d\phi = \frac{\gamma ds}{2\pi} \theta$$

Where γds is the total circulation of the segment **ds**.

- Add up the contribution from all the segments on **p**,

$$\phi(x, y) = -\frac{1}{2\pi} \int_a^b \theta \gamma ds$$

$\Gamma = \int_a^b \gamma ds$ - Total circulation integrated along an arbitrary curve between points **a** and **b**.

- Discretize the sheet into panels with boundary points and control points.
- The velocity potential on the i^{th} panel due to j^{th} panel is

$$\Delta\phi_j = -\frac{1}{2\pi} \int_j \theta_{ij} \gamma_j ds_j$$

- The total velocity potential on the i^{th} panel due to all the panels is given by

$$\phi(x_i, y_i) = -\sum_{j=1}^n \frac{\gamma_j}{2\pi} \int_j \theta_j ds_j$$

- Applying the boundary condition that the normal velocity at the surface must be equal to zero (**no penetration condition**).
 - Due to freestream velocity: $U_\infty \cos \beta_i$
 - To get the velocity due to the vortices, we take the spatial derivative of the local velocity potential in the direction normal to the surface.

$$\frac{\partial}{\partial n_i} \phi(x_i, y_i) = -\sum_{j=1}^n \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j$$

Hence,

$$U_\infty \cos \beta_i - \sum_{j=1}^n \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j = 0$$

This represents a set of n equations to be solved. This equation represents the primary equation for the vortex panel method.

- Finding the correct gamma distribution allows us to recreate the flow field successfully where the streamlines equal bodies.
- To get lift force,

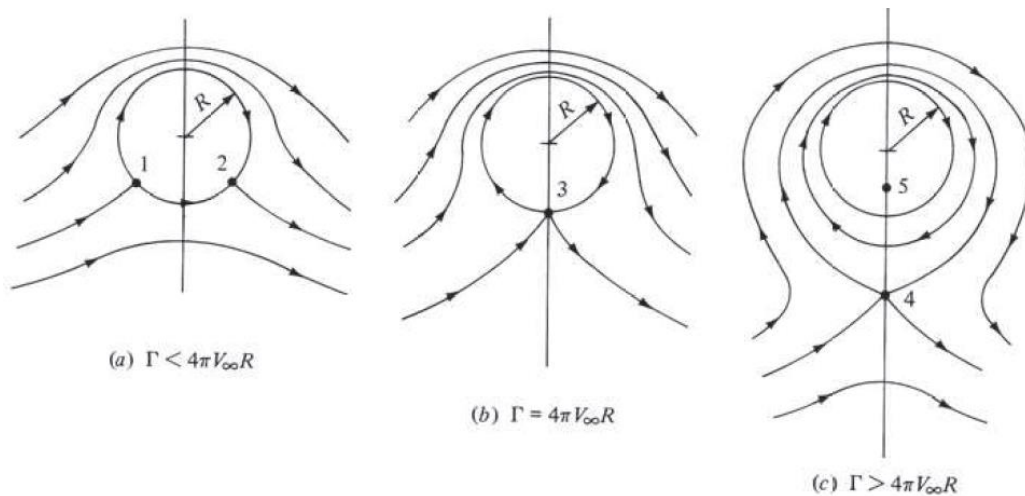
$$L' = \rho U_\infty \Gamma$$

- Recall from the rotating cylinder analysis that the strength of circulation parameter can influence the stagnation points on the body pushing them around the surface.
- We have to consider the effect of circulation on the stagnation points in the vortex panel method also.
- For non-lifting bodies, the stagnation points naturally occur at the leading and trailing edge of the body.
- For lifting bodies, the stagnation points unphysically reside on the top and bottom surface of the body based on the strength of circulation.
- To fix the stagnation points at the trailing edge, we separately enforce a condition called **Kutta condition**, which forces the stagnation point to be on the trailing edge or the point at the end of the airfoil.

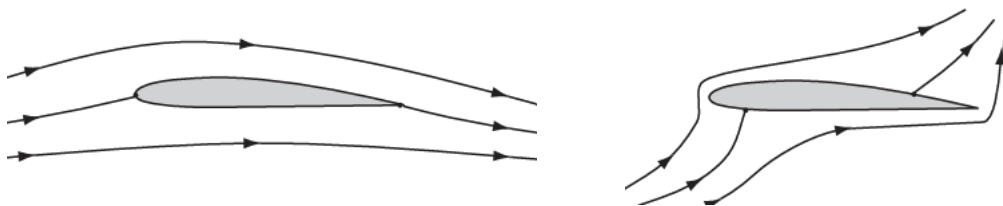
Kutta condition:

- When we model the flow around a cylinder using the potential flow approach. How many potential flow solutions are possible? **Answer: Infinitely many!**

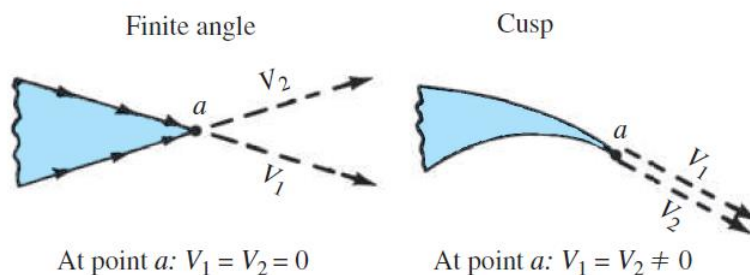
- Lifting flow over a circular cylinder – an infinite number of potential flow solutions were possible, corresponding to the infinite choice of vortex strength (Γ) as shown below Figure.



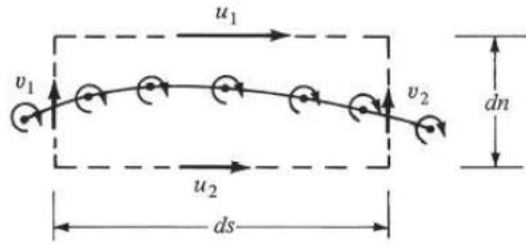
- The same situation applies to the potential flow over an airfoil.
- For a given airfoil at a given angle of attack, there are an infinite number of valid theoretical solutions, corresponding to an infinite choice of vortex strength (Γ).



- Although there is an infinite number of possible potential flow solutions, nature knows how to pick a particular solution.
- In a physical flow (**i.e. having a viscous effect**), the flow will smoothly leave the top and the bottom surfaces of the airfoil at the trailing edge.
- In the steady flow over a given airfoil at a given angle of attack, nature adopts a particular value of circulation which results in the flow leaving smoothly at the trailing edge.
- Flow turning around a sharp corner has an infinite velocity at the corner for potential flow. **This is physically incorrect.**



Tangential velocity jumps across the vortex sheet:



From the definition of circulation, the circulation around the dashed path is

$$\Gamma = -(v_2 dn - u_1 ds - v_1 dn + u_2 ds)$$

$$\Gamma = (u_1 - u_2) ds + (v_1 - v_2) dn$$

We know that,

$$\Gamma = \gamma ds$$

Let $dn \rightarrow 0$, u_1 and u_2 become the velocity component tangential to the vortex sheet immediately above and below the sheet.

$$\Gamma = \gamma ds = (u_1 - u_2) ds \Rightarrow \gamma = u_1 - u_2$$

The above equation states that the local jump in tangential velocity across the vortex sheet is equal to the local sheet strength.

Trailing edge with a finite angle: Flow leaves the trailing edge smoothly when V_1 and V_2 must be equal to **zero** at point **a**.

$\gamma(\text{T.E.}) = V_1 - V_2 = 0$ **(the local jump in tangential velocity across the vortex sheet is equal to the local sheet strength.)**

For the cusped trailing edge: Flow leaves the trailing edge smoothly when V_1 and V_2 must be equal in magnitude and direction at point **a**.

$$\gamma(\text{T.E.}) = V_1 - V_2 = 0$$

If V_1 and V_2 are not equal in magnitude, then there will be a pressure jump at point **a** **(according to Bernoulli's equation)** and it cannot act as a stagnation point.

$$P + \frac{1}{2} \rho V^2 = \text{const}$$

We can summarize the statement of the **Kutta condition** as follows:

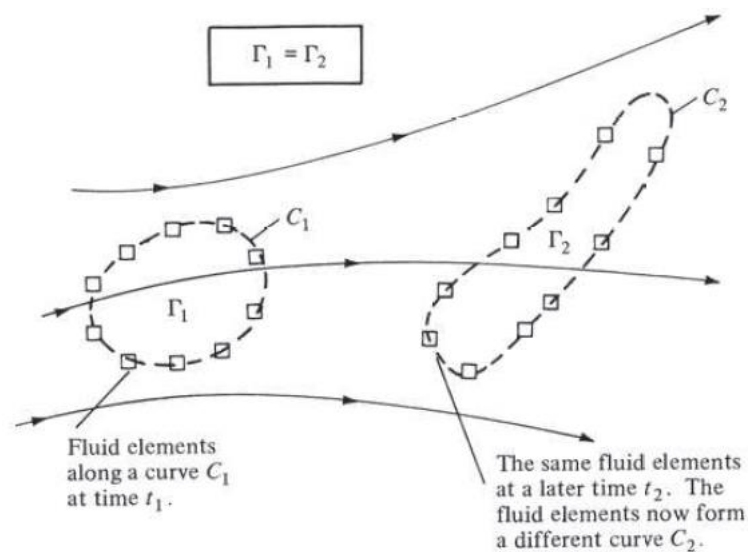
1. For a given airfoil at a given angle of attack, the value of Γ around the airfoil is such that the flow leaves the trailing edge smoothly.
2. If the trailing-edge angle is finite, then the trailing edge is a stagnation point.

- If the trailing edge is cusped, then the velocities leaving the top and bottom surfaces at the trailing edge are finite and equal in magnitude and direction.

Kelvin's Circulation Theorem (**Conservation of circulation**):

The time rate of change of circulation around a closed curve consisting of the same fluid elements is zero.

$$\frac{D\Gamma}{Dt} = 0$$



$V = 0$ $\Gamma_1 = 0$	$\Gamma_2 = \Gamma_1 = 0$ $\Gamma_3 = -\Gamma_4$
Consider an airfoil initially at rest. Since $V = 0$, $\Gamma_1 = 0$ We know that $\Gamma = \int_a^b \gamma ds$	Flow over the airfoil with starting vortex: Since $V \neq 0$ leads to a non-zero circulation $\Gamma_4 \neq 0$ Experimental flow observation shows that a starting vortex of circulation Γ_3 is shed from the trailing edge. $\Gamma_2 = \Gamma_3 + \Gamma_4$ $\Gamma_2 = 0$ (According to Kelvin's theorem)

	If $\Gamma_2 = 0$, then $\Gamma_3 = -\Gamma_4$ Therefore, the airfoil and the starting vortex must have equal and opposite circulation.
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Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.